



EDUCATION				
Program	Degree	Institution	% / CGPA	Year
Dual Degree (B.Tech + M.Tech)	Engineering Design, <i>Specialization - Robotics</i>	Indian Institute of Technology Madras	8.3/10	2024
Class XII	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	95.4%	2019
Class X	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	10/10	2017
- Awarded (among 30 out of 800+ students) the IIT Madras Young Research Fellowship based on remarkable research potential. 2021 - Secured 97.6457 Percentile in JEE Advanced and 99.1046 Percentile in JEE Mains out of over 1.5 Million aspirants across India. 2019 - Secured an All India Rank of 51 in UCEED(Undergraduate Common Entrance Examination for Design) conducted by IIT Bombay. 2019				
RELEVANT COURSES AND SKILLS				
Courses	Deep Learning, Field and Service robotics, Control Systems, Design of Mechanical Systems, Design of Thermal and Fluid Systems, Operation Research, Fundamentals of Automotive System, Mechatronics System Design.			
MOOCs	Algorithmic Toolbox*, Introduction to Reinforcement Learning*, Game Development, Introduction to Tensorflow			
Skills	Languages: Python, C, C++, C#	Tools: VS Code, MATLAB, LaTeX, MS-Office, Unity3D, Fusion360, Linux		
	Frameworks: Tensorflow, Pytorch	Libraries: Pandas, NumPy, Scikit Learn, Matplotlib, OpenCV, PyQt5, OpenGL		
RESEARCH AND PROFESSIONAL EXPERIENCE				
Yaskawa Pvt. Ltd <i>Robotics Intern</i> Jun 2022 – August 2022	Yaskawa Electric Pvt. Ltd. is an industry leader in Arc welding, Spot welding, Handling, Palletizing and Painting robots.			
	- Modeled and analyzed three-dimensional robotics palletizing algorithm and mechanics for mixed carton sizes. - Programmed and simulated arc welding and palletizing robots using MotoSim EG-VRC^[10] and teach pendant. - Developed a C++ algorithm to effectively use robot's memory when sending data to the robot from computer.			
Young Research Fellow IIT Madras Nov 2022 – Present	An illustrious program funded by the IITM batch of '76 to support undergraduate research; Guide - Prof. Satadal Ghosh			
	- Developed a 10% faster hybrid algorithm incorporating online obstacle avoidance & path planning algorithms. - Simulated it in MATLAB environment; Drafting a research paper, to be submitted in IEEE Aerospace Journals			
Blurgs <i>Computer vision Intern</i> Jun 2021 – Aug 2021	Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action			
	- Analyzed and tested 7+ State of the art Multiple Object Tracking networks for the startup's first product. - Trained and tested various MOT networks like Siam-MOT and FairMOT for Surveillance with Drones and CCTV. - Used FairMOT to track and get the trail of multiple moving objects and Re-identify objects in real-time.			
Avishkar Hyperloop <i>Power Systems Engineer</i> Oct 2020 – Jun 2021	The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country			
	Winner of the most scalable design award by Zeleros¹ and among the top 5 for mechanical and propulsion. - Designed a batter pack with high discharge rate for higher speeds while ensuring safe power supply to the pod - Worked on a custom-made Battery Management System to maintain voltage, temperature and ensure safety. - Extended battery pack life by 30% by reducing peak operational temperature by 10°C using an Al⁴ heat sink.			
PROJECTS				
CFD ⁷ Project (Mar 2021 - Apr 2021)	- Designed an aero foil for the Front wing of a Formula Student car for maximum Downforce/Drag ratio. - Improved Force/Drag ratio by 33% by varying position and angle of the aero foil relative to ground and car. - Performed CFD⁷ analysis to determine maximum possible ratio and the optimal setup for better efficiency.			
	- Lead a 6-member team involved in ideation of a novel mechanism of a fully mechanical alternative of a Segway - Improved efficiency by 50% by replacing conventional circular-gear pedaling system with rectilinear pedaling. - Carried out kinematic analysis to determine 6+ parameters using MATLAB and SIMULINK to improve efficiency			
Human Powered Segway (Apr 2021 – Jun 2021)	- Designed a small robotic arm with 2 degree of freedom for identifying and lifting objects using electromagnets - Programmed using Arduino mega and used proximity sensors for detection and LED light for sensing color			
Robotic Arm (Apr 2021 – Jun 2021)				
POSITION OF RESPONSIBILITIES				
Placement & Internship Cell	Department Core (May' 22 - Present)	Heading & mentoring a 3-tier team in charge of department placement and internship process 1200+ students - Analyzed past 3 years' list of contacts to eliminate redundant calling in order to increase company conversion. Planning & executing the logistic framework for the first ever shift to hybrid mode of campus recruitment.		
	IPR Coordinator (May'21 – Apr'22)	- Member of team managing 200+ non-engineering profiles in sectors such as consulting, finance and analytics. - Discovered and contacted 20+ potential recruiters based on skill-set requirements and student requests.		
CVI ⁸ Club, Coordinator (May'20- Apr'21)		- Responsible for the club activities like conducting peer-to-peer sessions, mini-projects, events and projects. - Organized and conducted sessions where basics of OpenCV, Machine Learning & Deep Learning were taught.		
Coding and Logic Events Shaastra'21 ^[3]		- Handled a total footfall of 300+ teams and crafted a total of 50+ logical questions for events in Shaastra'21. Ideated and executed 2 brand new events, Mystery Rooms and E-Contest with 100+ teams participating.		
EXTRA CURRICULAR ACTIVITIES				
Drawing		Won 5+ state level competitions and 10+ District level competitions in drawing (Oil Pastels and Pencils)		
Teaching Assistant		Assisted professors in coursework and grading of 20+ students in the courses LifeSkill and Happiness&Success.		
Game Dev. / Gaming		- Love to try out various tools and features in Unity3D; Reached a rank of Silver 2 in Valorant by Riot Games. - Solely developed a complete 3D Game in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.		
Sports		Among the 20 out of 100 students selected for freshman Volleyball training; Part of the Hostel Volleyball team		
Fusion360 API		Developed a python script to automatically create models in Fusion360 just by specifying dimensions.		

[1] Zeleros – A Spanish company that develops Hyperloop; [2] Virtual Reality; [3] Application Programming Interface; [4] State of the Art; [5] User Interface; [6] Tool for designing and building graphical user interfaces; [7] Computational Fluid Dynamics; [8] Computer Vision and Intelligence Club; [9] ISO Certified Annual Technical Festival of IIT Madras; [10]; Enhanced Graphics Virtual Robot Controller